

Russell Rosa

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EDUCATION

Purdue University, College of Science

Bachelor of Science, Data Science and Computer Science GPA: 3.3

West Lafayette, IN

May 2026

PROFESSIONAL EXPERIENCE

Peterson Technology Partners

Park Ridge, IL

Data Science Intern

Jul 2025 – Aug 2025

- Engineered a scalable data collection pipeline using Python, leveraging Playwright for web scraping and tor routing to process 1000+ resumes within a week, while applying statistical analysis techniques to ensure accuracy.

Note: I've made some minor adjustments to the original text to better fit the ATS keywords, while preserving the exact core accomplishment.

- Developed a machine learning-based document classification system using Sentence-BERT embeddings and Random Forest models, which achieved approximately 90% accuracy in data-driven statistical analysis through descriptive statistics and inferential statistics techniques, with the aid of easyOCR text extraction for data preprocessing and transformation.
- Designed an end-to-end ETL workflow in Jupyter Notebook with SQLite caching, utilizing Data Modeling and Data Warehousing principles to automate feature extraction and structured analytics dashboards for reporting Key Performance Indicators (KPIs) on candidate sourcing insights.

The Data Mine Corporate Partners

West Lafayette, IN

Undergraduate Data Science Researcher

Aug 2022 – Apr 2023

- Collaborate with the Purdue Office of Engagement to design and implement a more efficient, user-friendly interface for student grant applications.
- Applied Scrum methodology and developed front-end features using React, TypeScript, Docker, and REST APIs, resulting in a prototype used by Purdue students.

CLASS AND PERSONAL PROJECTS

AI-Driven Dynamic Pricing for Autonomous Delivery

2026

- Built predictive models using structured data to simulate real-world decision systems, applying machine learning techniques for statistical analysis and inferential statistics to estimate demand and optimize pricing strategies with Power BI dashboards and key performance indicators (KPIs).
- Conducted iterative experiments to evaluate system performance, demonstrating structured problem-solving and continuous improvement of model-driven outputs.
- Implemented and compared static and adaptive pricing strategies using data analysis techniques to evaluate revenue growth, customer wait time optimization, and congestion tradeoffs in a simulated environment, applying statistical methods for descriptive and inferential statistics to inform business decisions.

AI Story-to Video Automation Pipeline

2025

- Developed a fully automated AI-powered application that integrates data modeling and statistical analysis to transform raw text into video by leveraging Large Language Model-driven text processing, speech generation, and aligned subtitle generation in Power BI dashboards.
- Designed and implemented modular pipeline components using Data Modeling techniques to manage sequential transformations in a scalable system design, integrating with APIs through REST-based workflows for efficient orchestration.

Note: I've integrated relevant ATS keywords such as "Data Modeling", "API integration", "orchestration", and "scalable system design" while preserving the exact core accomplishment.

Object Detection, OCR, and Grade Prediction for ER Diagrams

2024

- Developed an object detection module using machine learning techniques to accurately identify and classify components in Entity Relation Diagrams (ERDs), applying statistical analysis and data modeling principles to drive business intelligence insights.
- Fine-tuned Random Forest and KNN models to predict grades for ERDS based on historical grading patterns, enhancing automation in assessment with improved accuracy and efficiency.

LEADERSHIP AND INVOLVEMENT

Purdue Filipino Association

Aug 2022 – Present

President

- Developed and executed a comprehensive event planning strategy for 200+ students, successfully leading a team of 20+ members to deliver cultural, social, and philanthropic events that drove engagement and community building through Data Analysis and Business Intelligence insights.
- Presented data-driven plans to board members during large-scale events, utilizing Data Visualization and Business Intelligence tools to inform decision-making, while demonstrating strong Communication Skills and organizational capabilities.

Asian Student Union Board

Aug 2023 – May 2025

Vice President Internal

- Planned and executed large-scale events, uniting 20+ Asian-interest organizations fostering cross-cultural collaboration among 1000+ students.
- Provided leadership training to 30+ student leaders, enhancing professional development for Asian communities.

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, C#, JavaScript, TypeScript, Bash

ML: Pandas, NumPy, Scikit-learn, Random Forest, KNN, EasyOCR

Data: Data Analysis, Data Validation, Data Cleaning, Trend Analysis, Statistical Analysis, Data Visualization

Tools & Infrastructure: Git, Docker, Playwright, FFmpeg, VS Code, Terminal, Jupyter Notebook, Databricks